SCST: An Enterprise-Class SCSI Target Framework for Linux

SCST is an SCSI target software framework. The objectives of this framework are high performance, high reliability and strict conformity to existing SCSI standards. SCST supports multiple SCSI protocols and local storage interfaces.

SCST is a generic SCSI sub-system for Linux that allows the creation of sophisticated storage devices from any Linux machine. The storage devices created can provide various functionalities like SCSI-3 persistent reservation, replication, thin provisioning and a range of other functions which normal storage devices offer. In this section, we will discuss the iSCSI feature of SCST on Linux machines.

Getting familiar with the terminology used

Before going forward, let's get familiar with some terms commonly used in the storage domain and in this article.

SCSI (Small Computer System Interface): This is a set of standards for physically connecting and transferring data between computers and peripheral devices. Some examples of SCSI devices are SCSI hard drives, SCSI CD-ROM drives and SCSI tape drives.

iSCSI (Internet Small Computer System Interface): This works on top of TCP (Transport Control Protocol), allowing you to send SCSI commands in a local area network (LANs). iSCSI is the way by which the user can share block level storage over a network, unlike NFS (Network File System), which shares storage at the file system level.

iSCSI target (server): iSCSI infrastructure is a client-server architecture. The server which provides storage resources is called the 'target'. iSCSI's target serves storage and other services to clients, and also takes care of SCSI commands received over the network.

iSCSI initiator (client): This is basically a client in the iSCSI infrastructure. Almost all mainstream operating systems, by default, provide an iSCSI initiator package. An initiator sends SCSI commands to the iSCSI target over the network in order to get services and storage served.

Configuring SCST in Linux

On the target system

SCST can be installed from the source available online at *https://svn.code.sf.net/p/scst/svn/trunk/* but before installing it, you need to install kernel development packages for your system.

On Debian systems: To install on Debian systems, give the